

Analysis of Heart Rate Variability using PPG signals to detect and predict emotions through Machine Learning.

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Abstract

This project focuses on developing a machine learning pipeline that predicts emotional arousal levels based on HRV features derived from Photoplethysmography (PPG) signals. Data was collected from participants while they were exposed to emotion eliciting video stimuli, capturing their physiological responses via PPG sensors. The pipeline involves several key stages: signal preprocessing (including Butterworth filtering), extraction of time-domain, frequency-domain, and non-linear HRV features from Poincaré plots, normalization using a stage of the experiment for features calibration, and relation analysis to assess feature relevance. The processed data was then used to train and evaluate regression models such as XGBoost, Random Forests, Support Vector Machine Regressors, validating with Leave-One-Out Cross-Validation (LOOCV) method to predict arousal levels.

Evaluation metrics like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), R-squared (R^2) were used to assess model performance. The study demonstrates that HRV features extracted from PPG signals could serve as reliable predictors of emotional arousal, offering a non-invasive and scalable solution for applications in neuromarketing, healthcare, and human-computer interaction. That is said after achieving a strong arousal predictive result with a Random Forest Regressor of a value correlation of 0.5975 between the predicted values and the true values, a Root Mean Squared Error of 0.154 ± 0.032 , a Mean Absolute Error of 0.122 ± 0.033 and a P-value of $7.9171e-199$. Without doubt the performance and the results of the Machine Learning models depend clearly on the data, meaning that the data extraction is as important as are all the other pipeline steps, methods and all the techniques that lead to the data ready for the prediction procedures, thus, it is indispensable to obtain it and preprocess it as best as possible to obtain strong and even very reliable results.

Keywords: Heart Rate Variability (HRV); Photoplethysmography (PPG); Emotion; Recognition; Arousal Detection; Machine Learning; Poincaré Plot; Neuromarketing; Support; Vector Regression (SVR); Linear Regression; Biosignals.

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Introduction

Neuromarketing is a growing field that integrates neuroscience, psychology and artificial intelligence to understand people's behaviours and use that to target strategies more effectively and get better outcomes at the moment of promoting different types of things. Traditional marketing approaches, such as surveys, focus groups, data collection and analysis, often fail due to the lack of honesty, capacity to distinct emotions, among others. Neuromarketing seeks to approach this kind of problem by trying to give more precise analysis since it involves analysing biometric data, including brain activity, eye movements, skin conductance, and heart rate variability, following the purpose of providing objective insights into people's emotions [1]. People's emotions insights through biosignals are highly important because they show how people honestly react to different stimuli.

These studies involve different approaches, focusing on analyzing people's variations in their biosignals to comprehend their way of feeling. In order to analyze this, the biosignals are taken and then analyzed by valence and arousal. The valence represents if the emotion is positive or negative, meanwhile the arousal represents how strong the feeling is.

Among the various physiological signals used in emotion recognition such as EEG (electroencephalogram), GSR (galvanic skin response), and eye tracking, Heart Rate Variability (HRV) has gained significant attention due to its direct link to the autonomic nervous system (ANS). HRV measures the variation in time intervals between consecutive heartbeats (RR intervals) and reflects the balance between sympathetic and parasympathetic nervous activities. In other words, the balance of the nervous system which is best known for its role in responding to dangerous or stressful situations[2] and the nervous system which predominates in quiet “rest and digest” conditions [3]. Therefore, analysing the behaviour in HRV patterns can indicate emotional arousal, making it a valuable proxy for understanding affective states.

Photoplethysmography (PPG) is a non-invasive optical technique that measures blood volume changes in the microvascular bed of tissue. This sensor is based on the emission of infrared light by a LED which penetrates the skin and blood vessels. This light is captured by the detector to measure the blood stream. The results of the PPG signal depend primarily on the flow of blood and oxygen to the capillary vessels in each heartbeat [4]. These sensors, commonly found in wearable devices and pulse oximeters, offer a cost effective and scalable alternative to electrocardiograms (ECG) for deriving HRV metrics. However, PPG signals are susceptible to noise, motion artifacts, and baseline drifts, necessitating robust preprocessing techniques to ensure data reliability [5].

This project focuses on developing a comprehensive machine learning pipeline that predicts emotional arousal levels based on HRV features extracted from PPG signals. Participants were exposed to a series of audiovisual stimuli designed to elicit different emotional responses while their physiological data was recorded. The methodology encompasses signal preprocessing, feature extraction (including time-domain, frequency-domain, and non-linear features such as Poincaré plots), normalization using a stage of the experiment for features calibration, and relation analysis to assess feature relevance. Finally, regression models were trained to predict arousal levels, aiming to provide an accurate, non-invasive method for emotion recognition.

The primary goal of this study is to demonstrate that HRV features derived from PPG signals can be effectively used to predict emotional arousal through machine learning models. This approach has potential applications not only in neuromarketing but also in healthcare, human-computer interaction, and affective computing.

Research Questions

1. Which HRV features derived from PPG signals are most relevant for predicting emotional arousal?
2. How does the heart rate dynamically respond to emotional stimuli, and how does this reflect in HRV analysis?
3. What is the predictive performance of machine learning models (XGBoost, Support Vector Regression, Random Forest) in estimating arousal levels based on HRV features?

Contributions

- Development of a robust preprocessing pipeline for PPG signals, addressing noise and baseline drift issues.
- Extraction of comprehensive HRV features, including time-domain, frequency-domain, and non-linear metrics.
- Implementation of normalization techniques to standardize feature values across participants.
- Correlation analysis to identify the most predictive HRV features for arousal estimation.
- Comparative evaluation of regression models for continuous arousal prediction.

This thesis presents a systematic approach to emotion recognition using affordable and scalable PPG devices, contributing to the growing field of affective computing[6] with practical applications in both commercial and clinical settings.

Literature Review

Emotion recognition using physiological signals has gained high importance in recent years due its potential application in human technology interaction, healthcare and neuromarketing. While traditional approaches often rely on people's actions, words, reactions and interactions, these methods can be consciously controlled or faked. As a result, new analyses which focus more on involuntary physiological signals are being carried on offering a more objective and reliable approach to detecting emotional states.

Heart Rate Variability in Emotion Recognition

Heart Rate Variability has been widely recognized as an important biomarker to analyze the autonomic nervous system's (ANS) which is the one component of the peripheral nervous system that regulates involuntary physiologic processes [7]. This component is constructed of three divisions, which are the following: Sympathetic, Parasympathetic and Enteric. Talking about the first one, when it activates, it leads to a state of overall elevated activity and attention: the common response known as “fight or flight”. In such a process, the blood pressure and heart rate increases. On the other hand, the Parasympathetic nervous system makes reference to the “rest and digest” processes, meaning that the heart rate and blood pressure go lower [8]. Lastly, the Enteric nervous system is the one mainly in charge of regulating digestion and gut function, therefore is a nervous system which will not take relevance in this report, as the focus is in the HR response at different audiovisual stimuli.

In context of emotions, it is expected that high arousal emotions, such as stress or excitement, typically result in decreased HRV, since there will be more consecutive heart beats in a shorter period of time, while relaxed states are associated with higher HRV values, since there is more chance to observe variation as the person is not in a consecutive intense way of feeling.

PPG as a tool for HRV analysis

Electrocardiography (ECG) is considered as the gold standard for HRV analysis, since it can be more precise and accurate. However, ECG setups can be intrusive and uncomfortable, making them less practical for robustness, real time or large scale applications. Photoplethysmography (PPG), on the other hand, provides a non-invasive and cost-effective alternative. PPG sensors, which are often embedded in wearable devices, measure blood volume changes, being able to derive Inter-Beat Intervals (IBIs) for HRV analysis[9].

Multiple studies have explored the use of HRV features derived from PPG signals for emotion recognition. Raj Rakshit et al. (2016)[10] proposed an emotion recognition system using time domain and frequency domain features extracted from PPG signals obtained via a fingertip pulse oximeter. They utilized Support Vector Machines (SVM) to classify emotions through video stimuli, focusing on three categories: happy, sad, and neutral. Their system achieved an average recognition accuracy of 83.8%, demonstrating the feasibility of using low cost and non-invasive sensors in emotion detection [10].

Expanding on the use of PPG signals, Goshvarpour and Goshvarpour (2018) introduced a non-linear analysis approach employing Poincaré's section measures to characterize the phase

space geometry of PPG signals during emotional states. Their study classified emotions experienced during music videos into three categories: love, hate, and fun. By analyzing the geometric indices and statistical features derived from Poincaré's sections and applying SVM classifiers with various kernel functions, they achieved classification accuracies of 96.67% for binary and 91.11% for multi-class emotion recognition. This work highlighted the effectiveness of non-linear dynamics in capturing complex patterns in physiological signals related to emotional responses[11].

Regression Based Approaches

It can be said that many studies have been made, but it is difficult to find specific ones regarding only utilizing PPG to predict arousal magnitudes; however, important regression based related studies exist, for instance; Silva et al. (2022) presented a study on real time HRV estimation using only PPG, reporting an RMSE of 0.178 and a correlation coefficient of 0.401 when comparing PPG derived HRV parameters to ECG as ground truth [12]. These findings suggested that PPG can be used for reliable HRV estimation, nevertheless, they emphasize how the huge impact of the high noise levels and experimental conditions affect the prediction results. On the other hand, as for many other Machine Learning model/technique with more preprocessed data, better results can be achieved, as in Schidt et al.(2018)'s paper in which WESAD dataset is introduced, one that collects multimodal physiological data (including PPG) from participants exposed to stress inducement tasks, amusement, and relaxation conditions. Even though not the three states were tried to be induced in an audio-visual way, the amusement state was and the performance metrics showed strong results, showing that a Long Short Term Memory(LSTM) model to predict in base of PPG can get a Mean Squared Error of 1.07 and a coefficient of determination equal to 0.64 at in relation with arousal and 0.72 and 0.48 with Valence respectively. This study focuses also on seeing which physiological signals are important for predicting these factors, for which they found that sometimes not all the biosignals are needed. While the study achieved good average results in emotion predictions, it concluded highlighting individual variability and emphasizing that as said before, reliable models could be developed even when using only a subset of biosignals[13].

HRV Time, Frequency Domain-Poincaré Plots and non-linear features.

Heart Rate variability (HRV) refers to the variations in time intervals between consecutive heartbeats, known as RR intervals or IBI (Inter-Beat Interval). In emotion recognition, HRV plays an important role because from them an analysis of the behaviour to understand the autonomic nervous system can be done.

Time-Frequency domain.

As in Raj Rakshit work[10], in time domain, 10 features such as meanRR, medianRR, SDNN, SDANN, pNN50, NN50, RMSSD, SDNN, meanHR, stdHR can be computed. On the other hand, in frequency domain analysis, Welch's algorithm must be carried out for spectral analysis. "Welch's method computes an estimate of the power spectral density by dividing the data into overlapping segments, computing a modified periodogram for each segment and averaging the periodograms" [15]. For example, the power spectrum of the HRV signal can be divided into 3 bands, as, very low frequency VLF (0-0.04 Hz), low frequency LF (0.04-0.15 Hz) and high frequency HF (0.15-0.5 Hz) where LF and HF bands are related to

sympathetic and parasympathetic activities. So that, some features can be extracted from this, as, peakVLF, peakLF, peakHF, aVLF, aLF, aHF, aTotal, pVLF, pLF, pHF, nLF, nHF and LFHF.

Domain	# Features	Feature	Description
Time domain	1	meanNN	mean value of NN intervals
		medianNN	median value of NN intervals
		SDNN	standard deviation of NN intervals
		SDANN	standard deviation of mean NN intervals
		pNN50	percentage of successive NN intervals differing by greater than or equal to 50 ms
		NN50	total number of successive NN intervals differing by greater than or equal to 50 ms
		RMSSD	root mean square of successive NN differences
		SDNNi	mean of all the standard deviations of NN intervals for all windows
		meanHR	mean value of heart rates
		stdHR	standard deviation of heart rates
Frequency domain	3	peakVLF	highest power in VLF band
		peakLF	highest power in LF band
		peakHF	highest power in HF band
	4	aVLF	raw area of VLF band (ms^2)
		aLF	raw area of LF band (ms^2)
		aHF	raw area of HF band (ms^2)
		aTotal	total raw area of VLF, LF and HF bands
	3	pVLF	relative VLF area w.r.t to total area
		pLF	relative LF area w.r.t to total area
		pHF	relative HF area w.r.t to total area
	2	nLF	normalized LF area (i.e. $\text{LF}/(\text{LF}+\text{HF})$)

		nHF	normalized HF area (i.e. HF/(LF+HF))
		LFHF	ratio of LF and HF areas

Table 1.0 List of time domain and frequency domain HRV features [10].

Poincaré Plots

In addition to time and frequency domain features, non-linear methods can provide further insight into the dynamics of heart rate variability. One way to approach this is with Poincaré Plots, a graphical representation of the correlation between consecutive RR intervals. It is a scatter plot where each RR interval is plotted against the preceding one (RR_{n+1} vs. RR_n) [16]. These plots provide not only qualitative but also quantitative information about the autonomic regulation of the heart.

Poincaré Plots are highly valuable to its qualities, as allowing visualization of the actual HRV. In the plots different shapes or dispersions can indicate various states. For example, an elongated plot indicates greater variability, whereas a compact one represents less variability [17].

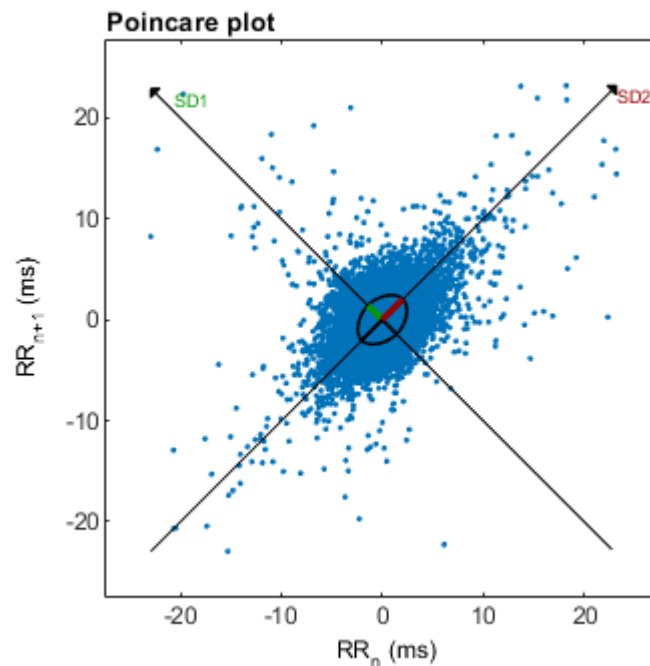


Fig 1.0 Example of visualization of a Poincaré Plots. [17]

In the study by Goshvarpour & Goshvarpour (2018) [11], Poincaré plots were extended beyond their typical use by being deeply analyzed, checking the geometric properties of its trajectories in response to emotional stimuli. By forming Poincaré Sections and extracting more features, they achieved high classification accuracy for different emotions.

As in the study by Goshvarpour & Goshvarpour (2018) [11] features that can be extracted from Poincaré Plots are:

Standard features such as SD1 and SD2. SD1 represents short term heart rate variability and reflects parasympathetic activity, while SD2 accounts for long term variability influenced by both branches of the autonomic nervous system. Additionally, the Parasympathetic Index, computed as the ratio $SD1/SD2$, provides insight into the balance between short and long term HRV dynamics.

Further features are derived from Poincaré Section Analysis, capturing more complex dynamics in heart rate behavior:

F1: Number of crossing points through Poincaré sections.

F2-F4: Minimum, maximum, and mean areas formed by normalized crossing points, representing variability and spread.

F5-F6: Standard deviations along the X and Y axes, representing dispersion.

F7-F8: Skewness of the X and Y coordinates, describing asymmetry in distribution.

F9-F10: Kurtosis of the X and Y coordinates, highlighting extreme values and outliers.

By extracting multiple, diverse and important features, later on, Feature Engineering can be done to select the best ones and then proceed to the Machine Learning part.

Meaning

- Time Domain HRV Features

Mean RR: The average duration between consecutive heartbeats. Lower values of Mean RR generally correspond to higher heart rates, which can occur in states of high arousal or stress. A shorter Mean RR interval may reflect increased sympathetic activation.

Median RR: Similar to Mean RR, but more robust against outliers. Like Mean RR, lower Median RR may be associated with higher emotional arousal or heightened alertness.

SDNN: Standard deviation of all RR intervals, representing overall HRV. Lower SDNN is often observed in emotionally aroused states such as stress or fear, reflecting reduced variability and autonomic inflexibility.

RMSSD: The root mean square of successive RR interval differences. This feature is sensitive to parasympathetic (vagal) activity and is known to decrease during emotional arousal, stress, or anxiety, indicating a reduction in short term HRV.

NN50: Counts the number of successive RR intervals that differ by more than 50 milliseconds. A high NN50 indicates strong beat-to-beat variability and tends to be lower during arousal, reflecting a reduction in parasympathetic tone.

pNN50: The percentage of NN50 relative to the total number of RR intervals. It provides a normalized measure of vagal tone, and like RMSSD and NN50, it tends to decrease under emotional stress or arousal.

- Frequency Domain HRV Features

aVLF: Absolute power in the very low frequency band (0.00–0.04 Hz). This band reflects long-term regulation mechanisms such as thermoregulation or hormonal influences. While less directly linked to arousal, changes here may suggest chronic stress or fatigue.

aLF: Absolute power in the low frequency band (0.04–0.15 Hz). Represents both sympathetic and parasympathetic activity, but is often associated with sympathetic modulation. In states of heightened arousal, aLF may increase due to sympathetic activation.

aHF: Absolute power in the high frequency band (0.15–0.4 Hz), strongly linked to respiratory sinus arrhythmia and parasympathetic activity. A decrease in aHF is commonly observed during stress, fear, or emotional arousal, indicating vagal withdrawal.

aTotal: Total power across the VLF, LF, and HF bands. A reduction in total power may indicate overall autonomic withdrawal and reduced HRV, which is typical in states of emotional arousal.

pVLF: Relative contribution of VLF power to total power. Less directly associated with acute emotional changes but may reflect broader autonomic trends.

pLF: Proportion of LF power relative to total power. Elevated pLF can reflect increased sympathetic tone, and it often rises during aroused or stressed emotional states.

pHF: Proportion of HF power relative to total power. Lower pHF is associated with decreased parasympathetic modulation and is typically observed during emotional arousal.

LF/HF Ratio: A widely used indicator of sympathovagal balance. A higher LF/HF ratio suggests increased sympathetic dominance (arousal, stress), while a lower ratio suggests parasympathetic dominance (calm, relaxed states).

nLF: Normalized LF power calculated as $LF / (LF + HF)$. This metric emphasizes the relative sympathetic contribution in short term HRV and tends to increase during arousal.

nHF: Normalized HF power calculated as $HF / (LF + HF)$. Reflects parasympathetic modulation, and decreases in nHF are typically linked to emotional arousal or stress.

- Poincaré Plot Features

SD1: Reflects short-term variability in heart rate and is primarily influenced by parasympathetic activity. Lower SD1 values are associated with emotional arousal, stress, or anxiety due to vagal withdrawal.

SD2: Represents both short and long term variability and incorporates both sympathetic and parasympathetic influences. A decrease in SD2 may reflect a loss of complexity in HRV under high emotional load.

SD1/SD2 (Parasympathetic Index): The ratio of SD1 to SD2 provides insight into the balance of autonomic activity. A lower ratio typically indicates sympathetic dominance, as seen during emotional arousal or tension.

Advanced Poincaré-Derived Features (F1–F10)

F1_CrossingPoints: The number of times the Poincaré plot trajectory crosses directional lines. A higher number of crossing points suggests increased HRV complexity and variability. In emotionally aroused states, this count may decrease, reflecting more predictable cardiac patterns.

F2_MinArea: The smallest product of normalized x and y coordinates of crossing points. A smaller F2 indicates tighter clustering of points and reduced variability, typically observed during high arousal or stress.

F3_MaxArea: The largest product of normalized crossing point coordinates. A larger F3 implies wider HRV spread, which is reduced under aroused or anxious states.

F4_MeanArea: The average area covered by crossing points, offering a general estimate of HRV dispersion. Lower F4 values suggest diminished HRV complexity and potential emotional tension.

F5_StdX: Standard deviation of the horizontal (IBI_n) crossing coordinates. Reduced variability here may indicate more regular heart intervals under stress or arousal.

F6_StdY: Standard deviation of the vertical (IBI_{n+1}) crossing coordinates. Lower F6 values reflect reduced flexibility in future heart intervals, commonly linked to arousal.

F7_SkewX: Skewness of the horizontal coordinate distribution. Positive or negative skew may indicate a bias toward shorter or longer intervals, respectively, which can be altered by emotional state.

F8_SkewY: Skewness of the vertical coordinate distribution, indicating directional bias in future RR intervals. Changes in F8 may reflect shifts in autonomic regulation during emotional changes.

F9_KurtX: Kurtosis of the horizontal distribution. High values reflect sharp outliers and abrupt changes in HRV, which may increase under emotional volatility.

F10_KurtY: Kurtosis of the vertical distribution. A high F10 suggests strong peaks in HRV response, potentially signaling erratic heart behavior during intense emotional moments.

Valence Arousal Model

The valence arousal model, proposed by Russell (1980), is a framework methodology of two dimensions widely used in emotion research[18].It categorizes in two axes :

- Valence : Measures the positivity or negativity of the emotion, unpleasant to pleasant.
- Arousal : Indicates the intensity or activation level of the emotion, deactivation to activation.

In this model since Heart Rate is the biosignal used in this project and which is known to be more useful to predict arousal, the focus is on arousal.

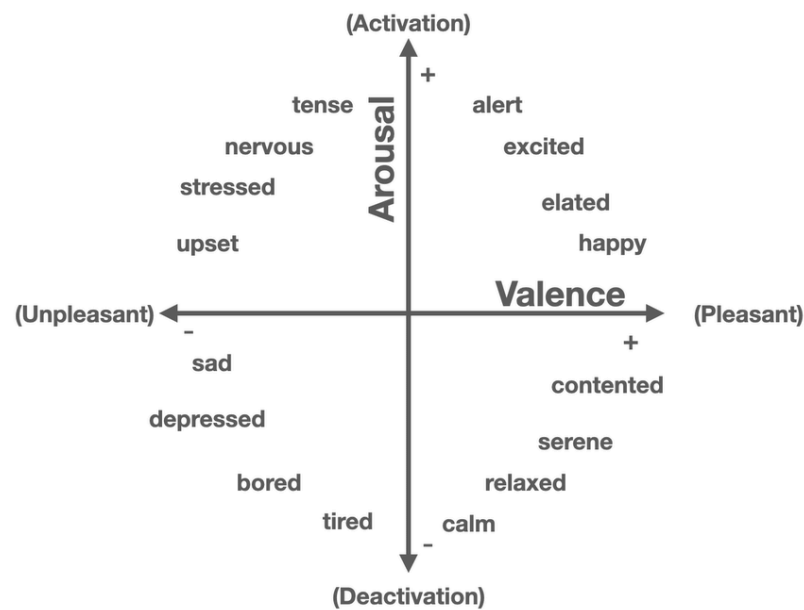


Fig 1.1 Example of Valence-Arousal framework [19]

Evaluation Methods

To evaluate the models performances in these projects, different metrics and validation techniques can be used. Since the goal is to predict emotion arousal levels based on the HRV features, the evaluation will focus on how well the models can make correlated predictions.

Metrics considered for the evaluation are:

- Mean Absolute Error (MAE): This metric shows how much the predictions are different from the actual values on average[20].

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

Equation 1.0 MAE

n is the number of predictions

y_i is the actual value

$\hat{y}_i$ is the predicted value

- Root Mean Square Error (RMSE): This metric measures the error but gives more importance to larger mistakes. It's useful to see if the model is making big errors in some cases[21]. This one measures the difference between the predicted values and the actual values, but it gives more weight to larger errors [21].

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

Equation 1.1 RMSE

- R-squared (R^2): This metric shows how well the model explains the variation in the actual data. If R^2 is close to 1, it means the model is doing a good job [22].

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Equation 1.2 R-squared

- Correlation: The Pearson correlation coefficient measures the strength and direction of the linear relationship between actual and predicted values [23]. Values close to +1 indicate a strong positive correlation, while values close to -1 indicate a strong negative correlation.

$$r = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}}$$

Equation 1.3 Correlation

- P-Value: The p-value associated with the correlation coefficient tests the null hypothesis that no correlation exists between predicted and actual values ($r=0$). A small p-value (typically < 0.05) indicates that the observed correlation is statistically significant[24].
- Concordance Correlation Coefficient (CCC): The CCC evaluates how well predictions conform to the ground truth by combining both precision (correlation) and accuracy (closeness to the 45° line through the origin). Unlike Pearson's r, it penalizes systematic bias between predictions and actual values.

$$CCC = \frac{2r\sigma_y\sigma_{\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2}$$

Equation 1.4 CCC formula

Where r is the Pearson correlation y_i , and $\hat{y}$ are means, and σ^2 , $\hat{\sigma}^2$ are variances of ground truth and predictions.

For validation, Leave-One-Out Cross-Validation (LOOCV) will be used. This works by training the model with all the participants' data, except for one, and then testing on that one participant. This process is repeated until every participant has been tested. LOOCV is a good choice because it helps make sure the model works well on new data, especially when there's not a huge amount of data to train on [20].

Gaps and Contributions of this work

For real world applications. Existing literature highlights the potential of HRV features derived from PPG signals for emotion recognition. However, there are still gaps to address:

- Many studies utilize multiple biosignals, complicating scalability
- Previous works often lack a thorough investigation of dynamic heart rate responses within short stimuli intervals.
- Regression based models for continuous arousal prediction using only PPG derived features remain underexplored.

This study contributes by developing a pipeline focused solely on PPG signals, extracting comprehensive HRV features, and leveraging machine learning regression models to predict continuous arousal levels. The approach emphasizes robust preprocessing, dynamic interval analysis, and normalization techniques to enhance model performance and reliability.

Methodology

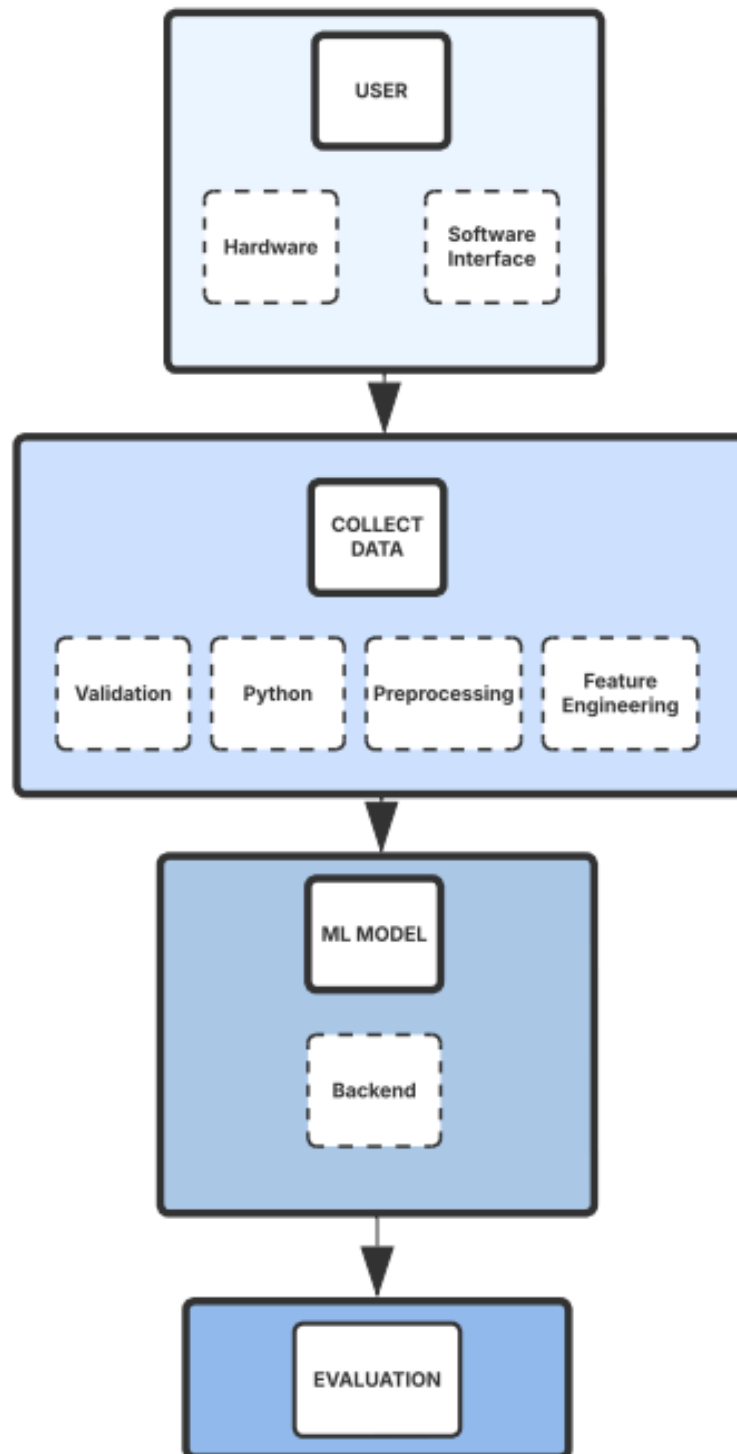


Fig 1.2 General flow of procedure.

1. Data collection

Participants were recruited to take part in an experiment where they were exposed to watch different video stimuli while multiple sensors were capturing their biosignals. The software linked to the hardware is called iMotions [26], which plays a crucial role in synchronizing and integrating all the biosignals collected from the sensors. This platform enables the seamless combination of multiple data inputs, including Eye Tracking, Galvanic Skin Response, Photoplethysmography signal, Electroencephalography, and Facial Expression Analysis. iMotions ensures that all these diverse data streams are accurately captured and processed, providing a thorough view of the participant's physiological and emotional responses during the experiment. At the end of the stimuli, participants answered a questionnaire meant to be used to get self reported emotional labels (valence and arousal). The purpose of these labels is to use them as the ground truth for training and evaluating the models.

For the procedures of this project 62 participants were considered, extracting their corresponding biosignals, the way the data is stored consists in the creation of a csv file per participant including the timestamps as all their physiological data. It is important to mention that for this project many more participants participated in the experiments, however, some of them were discarded due to many reasons that would affect negatively in the dataset, these reasons were: problems while recording the data or people with conditions such as depression and anxiety.

2. Signal Preprocessing

Once the PPG signals were collected, preprocessing had to be done to clean the data and enhance solid bases for further manipulation. As expected because of the sensors, much data was missing, so imputation data techniques had to be applied. Also, baseline drift was without doubts something to be treated since this can have irrelevant data or undesired content. In order to accomplish that, these signals must be processed by filter techniques which were applied in order to clean the data looking forward to finally obtaining the best data possible for the following procedures.

On the other hand, iMotions[26] helps to generate directly some features, such as Heart Rate, signal which was also used for the analysis, however after comparison, the work was fully done with PPG since it brought up better results. For this signal (Heart Rate feature from iMotions) no filtering techniques were required since the software already does the calculus, nevertheless, imputation data had to be done because as with the raw PPG signals some rows in the CSV's were missing.

Computational work for signal preprocessing: As mentioned before, the goal of this stage was to turn raw PPG into a clean, stimulus aligned dataset that would be ready for HRV feature extraction. Therefore, approaching that purpose a code was designed to follow the next pipeline:

First, the whole audio visual stimuli were analyzed.

1. Parsed raw CSVs and fixed headers

At the beginning of the CSVs files, unuseful data was present, that's why a function called `find_and_set_header` is applied which extracts the participants names for traceability and leaves a dataframe which starts from a reference cell which has the word "Row".

2. Selected columns and enforced numeric types

From the whole CSV only three besides the participant and the timestamps were essentials to keep:

- Internal ADC A13 PPG RAW (raw PPG)
- Heart rate PPG ALG (iMotions heart rate)
(Both signals were coerced to numerical).
- SourceStimuliName (labels of the current stimulus)

3. Handled missing values robustly

- For PPG: linear interpolation, then backward fill and forward fill as fallbacks (important when the gap occurs at the edges).
- For Heart Rate: first replace the invalid readings (-1) with NaN, then linear interpolation, then backward/forward fill so the series is continuous.

These imputation techniques are because PPG and HR sometimes have short dropouts; the order of the imputation techniques fills gaps without inventing extreme values. The way they work is the following: The linear interpolation is like drawing a straight line between the nearest known points[27], for example if the values are $t_1=707$ $t_2=\text{NaN}$ and $t_3=709$, then this technique would assign the value 708 to t_2 . On the other hand, the Backward fill(`bfill`) replaces missing values with the next valid observation. This is useful at the beginning of a gap, where no previous data exists. For instance, if a PPG signal is missing at the start of a recording, `bfill` can borrow from the next available point to avoid leaving empty spaces; while the Forward fill (`ffill`) works when the gap occurs at the end of a sequence or when continuity is important[28].

4. Filtered PPG with Butterworth filters.

- The sampling rate was 128 Hz.
- The baseline drift was removed with a 0.5 Hz high-pass Butterworth order 5 filter and then the high-frequency noise was removed with a 2Hz low-pass 5 order filter.
Both were applied with `filtfilt` (forward-backward), so no phase distortion was introduced.

This filtering technique because it keeps the fundamental cardiac component (and main morphology needed for peak detection) while removing drift and high frequency noise. `filtfilt` avoids time shifts that would harm RR estimation.

- The result was stored as “Butterworth Filtered PPG signal”.

5. Normalized timestamps per stimulus.

iMotions timestamps are global. The code converts the milliseconds to seconds and then reset each stimulus to start at 0 s, using group wise min subtraction. This was essential for future analysis and manipulation over the stimuli and across different people.

6. Standardize stimuli names

The code standardizes the names, shortening them for consistency and easier management.

3. IBI's plotted

After preprocessing the whole videos, the IBI's were plotted, this since watching the videos it could be clearly seen that some parts corresponded to high arousal moments and others to low arousal. In other words, cutting the videos and naming them differently happened to seem a good idea for better analysis. This idea was already applied when working with pupil size, then considered as the start marks for designing some intervals to try dedicated for the Heart Rate analysis. The IBI's plots were plotted for all the stimuli including 5 seconds from the surveys accordingly appending them so more data can be seen in the plots. The plots contain the delimitations of the pupil size intervals, precise biosignal which happen to have an immediate effect at reacting to different stimuli.

The plots look like these.

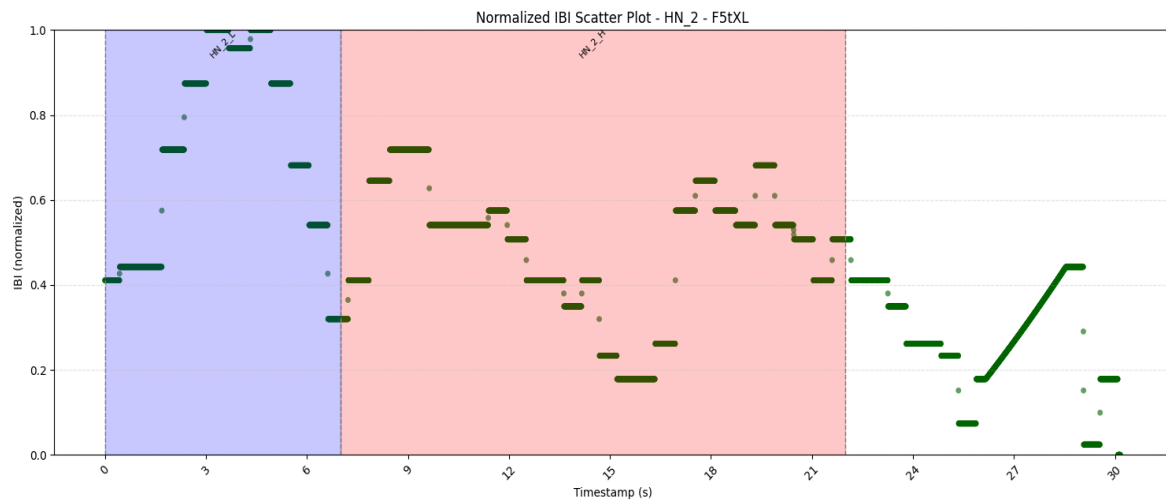


Figure 1.3 IBI scatter plot ex1

The blue area shows the low arousal part while the red one the high arousal part.

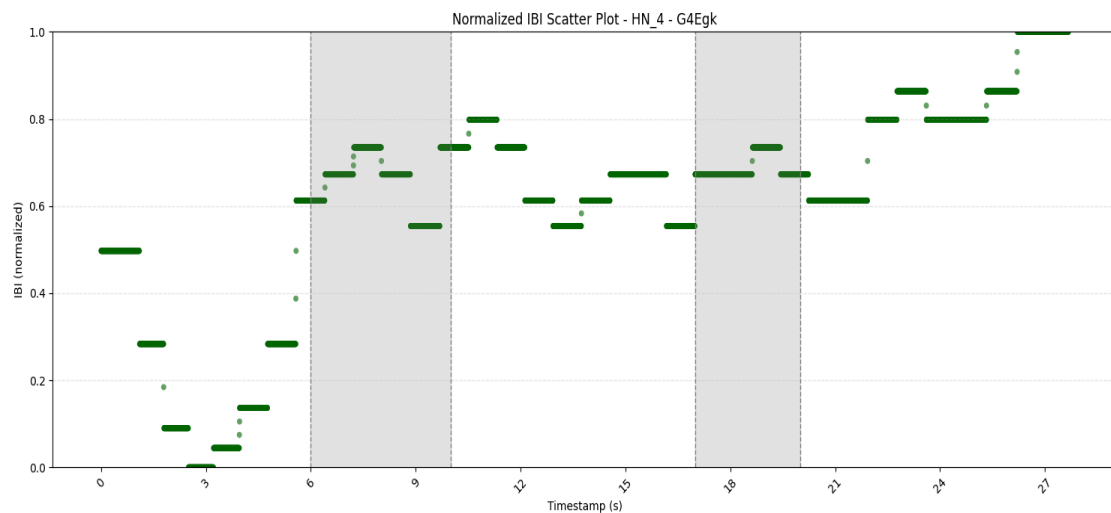


Figure 1.4 IBI scatter plot ex2

Other videos for pupil size did not have delimited low and high arousal parts, however they had intervals to merge them as one, looking forward to analyzing coinciding segments activating the same kind of response in the participants.

These plots, as mentioned before, represent the Interbeat Interval, the purpose was meant to plot dot by dot but when it looks like there are lines in the plots it is because of the imputation techniques.

These graphs show how the Interbeat Intervals change over time depending on the arousal of the participant. When arousal is high, the heart speeds up and the intervals between beats get shorter, so the dots are plotted lower on the y-axis and closer to the x-axis. On the other hand, when arousal is low the heart slows down, the intervals get longer, and the dots are plotted higher on the y-axis, further away from the x-axis. In simple words, points that appear lower in the plot are linked to higher arousal, while points higher in the plot are linked to calmer states.

Among all the audiovisual segments plotted, there were some which were focused on neutralizing the participants before, along and after the experiments. These were important to analyze to see which one would represent the most stable response from participants for further use in the normalization steps.

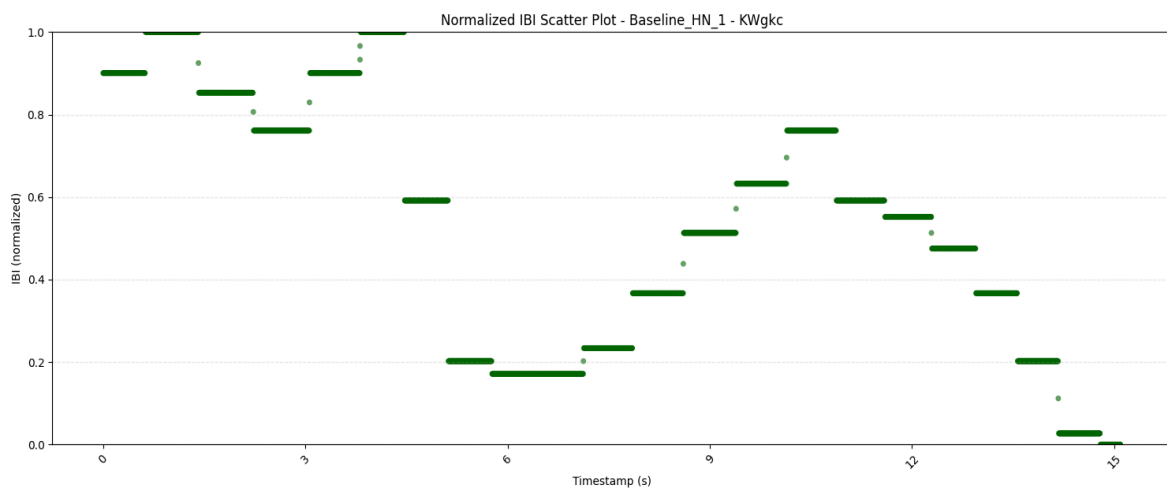


Figure 1.5 IBI scatter plot ex3
(Baseline), before every stimuli a baseline stimuli was played to neutralize the participant.

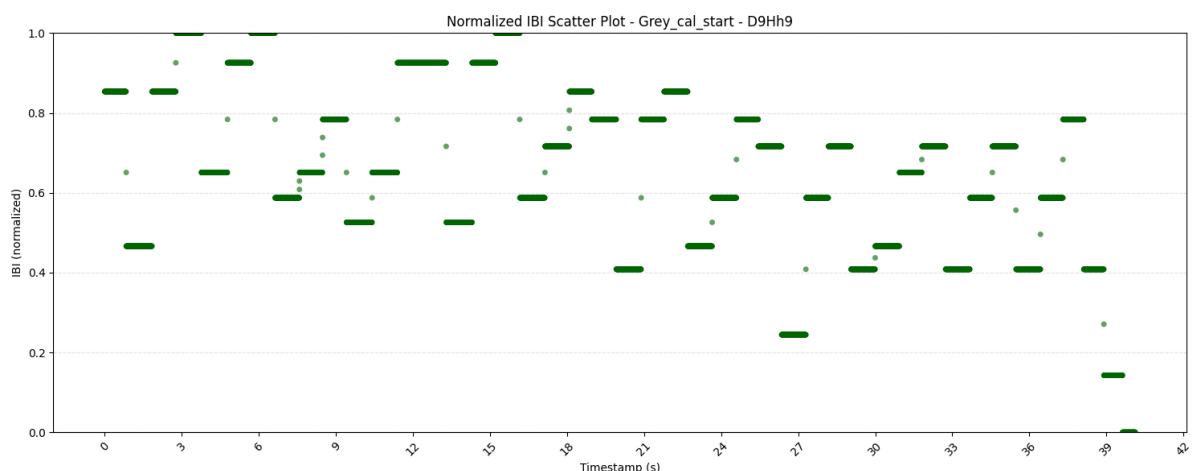


Figure 1.6 IBI scatter plot ex4
(Grey_cal_start), another stimulus to neutralize the participant.

As these two, some other stimuli were focused on neutralizing people's emotion and the plots were useful to analyze how well they did.

Then:

After analyzing the plots, it was clear that there was a lot of variability among each and every participant in their Heart Rate Variability performance. Moreover, as it could be assumed, the Heart Rate has not an immediate response as pupil size; thus, in order to proceed and procure the most accurate results, some adjustments had to be made.

After some research, several studies highlight the physiological delay of cardiovascular responses compared to other signals such as the pupil dilation. As mentioned before heart rate and its variability are regulated by the autonomic nervous system and require some seconds to reflect changes induced by emotional stimuli [27][28]. Therefore, keeping the original pupil based intervals would not adequately capture the delayed HRV dynamics, leading to misalignment between stimulus and the physiological response.

To address this, new intervals were created by shifting and extending the stimulus windows to account for the lag between audiovisual stimulation and HRV reaction. This adjustment allowed the extracted IBI segments to better synchronize with the actual heart rate reaction occurring during the emotional episodes, as suggested by previous works using PPG and ECG.

In practice, too many attempts were done to tackle this, first the analysis was made by amplifying and shifting the intervals 2 seconds, then by 3 seconds and even with 5 seconds. Considering these psychophysiological responses note a lag of 1–3 seconds for HR changes to align with emotional stimuli [28]. However, it is also mentioned that for analyzing Heart Rate Variability, it is important to consider as much time as possible, in order to understand the behaviours from beat to beat, therefore, another approach had to be taken to have more time per video interval and with it more IBIs per each of them.

As for HRV, GSR needs more time to be analyzed, for the same videos, some intervals called “whole videos” were designed, since most of the time of the stimuli were captured, getting longer intervals more useful for biosignals which take time to react. From those intervals, I proceed to design the intervals for Heart Rate (since trying with pupils’ intervals did not work), which are based on the start points of the GSR intervals but trying to extent the intervals to a minimum of 15 seconds were possible, meaning that if there was the need to shift them a bit or take time from the survey, it was done.

It is relevant to mention that the different intervals were evaluated in the following steps once the features were extracted and at the moment of seeing the relation they have with the ground truth, those were the moments of decision if proceed or to come back to this point to redesign the intervals. Already mentioned this, let's proceed with what happens once the intervals are selected.

4. Signal Preprocessing with intervals created

Once the intervals were created it was only a matter of separation the stimuli for better analysis; therefore, the same first steps were applied (handling missing data, filtering the

signals and normalizing the timestamps per stimulus), and from there the code is boosted with the following:

7. Split/merge special stimuli with precise intervals.

The videos were intentionally split into two parts with continuous time inside the stimulus., they were cut and renamed adding a letter according to the High or Low arousal part, for instance:

HP_1-1: keep[23 s, end] as HP_1-1, and merge [0, 23 s] as new video HP_1-2 (later renamed to high/low variant via the mapping to HP_1_L and HP_1_H)

8. Preserved baseline and calibration tools

- Baseline_ segments, End_Baseline, and the Grey calibration segments (Grey_cal_start, Grey_cal_mid) were kept, but treated separately from the emotional videos (they're used later for normalization and checks).
- Timestamps within each baseline block were also reset to start at 0 s.

9. Attached survey blocks to the correct videos

Short survey segments (e.g., survey_HP-7) were removed as standalone stimuli and appended at the end of their corresponding videos (e.g., HP_7_H), with timestamps shifted so time stays continuous in the target stimulus. This technique was to get at least 15 seconds per interval.

10. Final schema and export

- Only the following columns were kept to proceed with the further analysis. Respondent Name, Timestamp, SourceStimuliName, Internal ADC A13 PPG RAW, Butterworth Filtered PPG Signal, Heart Rate Signal.
- A final timestamp was run to reset per stimulus to ensure each block started at 0 s.
- The resulting per-participant file was written as filtered_<originalfilename>.csv to a fresh output folder.

5. Feature Extraction

After preprocessing the PPG signals and obtaining clean RR intervals, the next step involved extracting Heart Rate Variability (HRV) features. These features aim to capture the temporal, spectral, and non-linear dynamics of cardiac activity, which are linked to emotional responses. Following approaches in prior works, a diverse set of HRV features was computed.

Time-domain features:

Metrics such as Mean RR, SDNN (standard deviation of NN intervals), RMSSD (root mean square of successive differences), NN50 (number of successive RR intervals differing by more than 50 ms), and pNN50 (percentage of NN50) were extracted. These indices reflect the short- and long-term variability in heart rhythms, with SDNN representing global variability, while RMSSD and pNN50 are strongly associated with parasympathetic activity.

Frequency-domain features:

Using Welch's method for spectral analysis, the RR interval series was decomposed into its frequency components. Band-specific powers were calculated in the VLF (0.00–0.04 Hz), LF (0.04–0.15 Hz), and HF (0.15–0.40 Hz) ranges. Features included absolute powers (aVLF, aLF, aHF), relative powers (pVLF, pLF, pHF), normalized indices (nLF, nHF), and the LF/HF ratio. These indices are commonly interpreted as markers of sympathetic–parasympathetic balance.

In addition, Power Spectral Density (PSD) plots were generated for each participant stimulus pair to visually examine the distribution of spectral energy across frequency bands, supporting the interpretation of frequency-domain features.

Power Spectral Density (PSD plots)

These plots make it possible to visually confirm which bands dominate under different emotional stimuli. Stronger HF components generally indicate increased parasympathetic activation, while LF dominance is often associated with sympathetic modulation.

Examples.

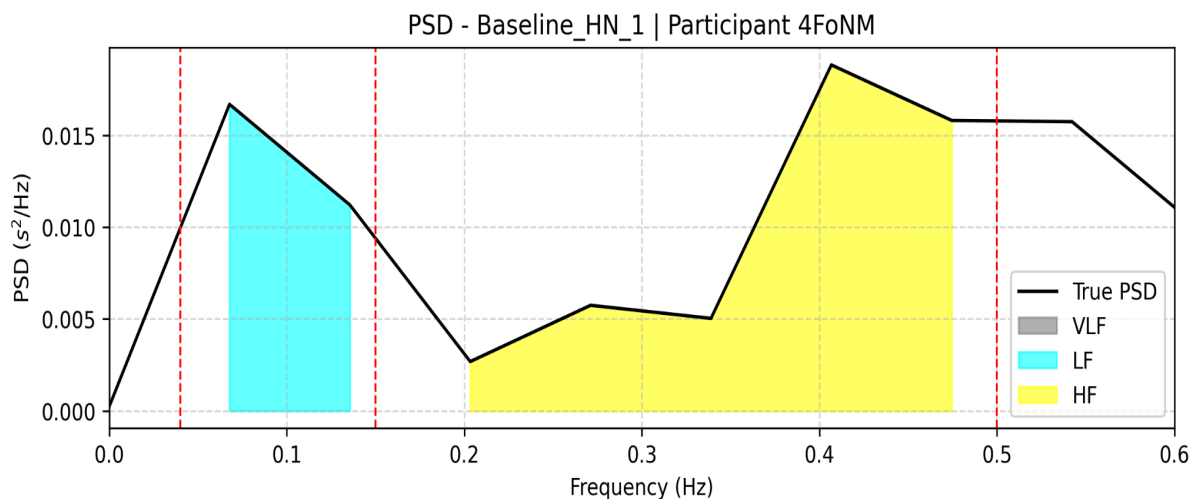


Figure 1.7 PSD plot Baseline_HN_1, Participant 4FoNM

While the participant 4FoNM was exposed to the baseline before video HN_1, the dominance was in the HF section, which happens to show a parasympathetic modulation, which makes sense since a phase to neutralize the participant.

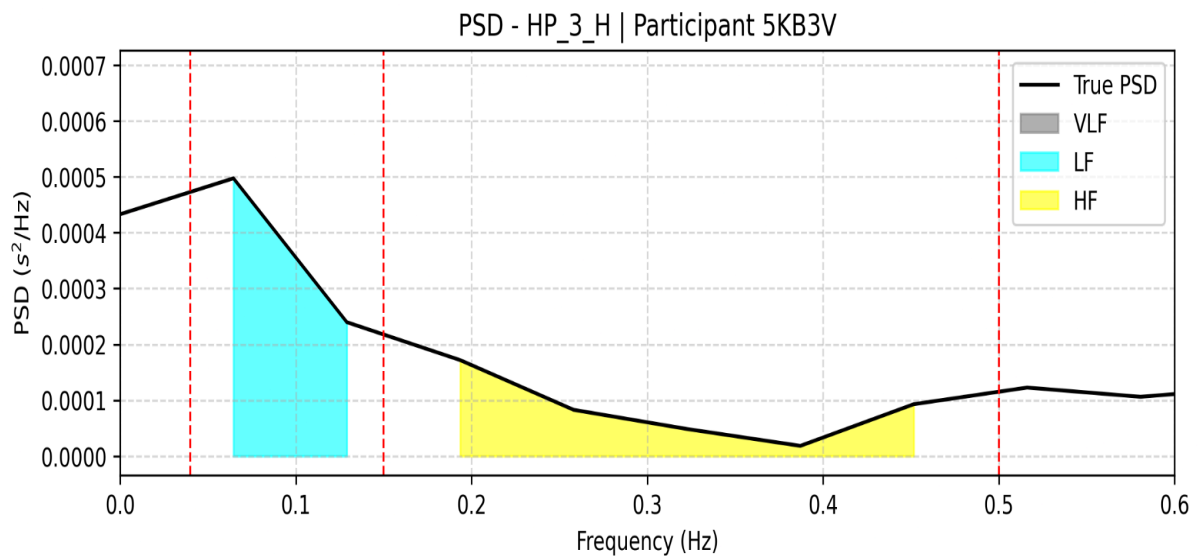


Figure 1.8 PSD plot HP_3_H, Participant 5KB3V

While the participant 5KB3V was exposed to the video HP_3_H, the dominance was in the section LF, which happens to show a sympathetic modulation, which makes sense since it is a video tagged with an H of High Arousal.

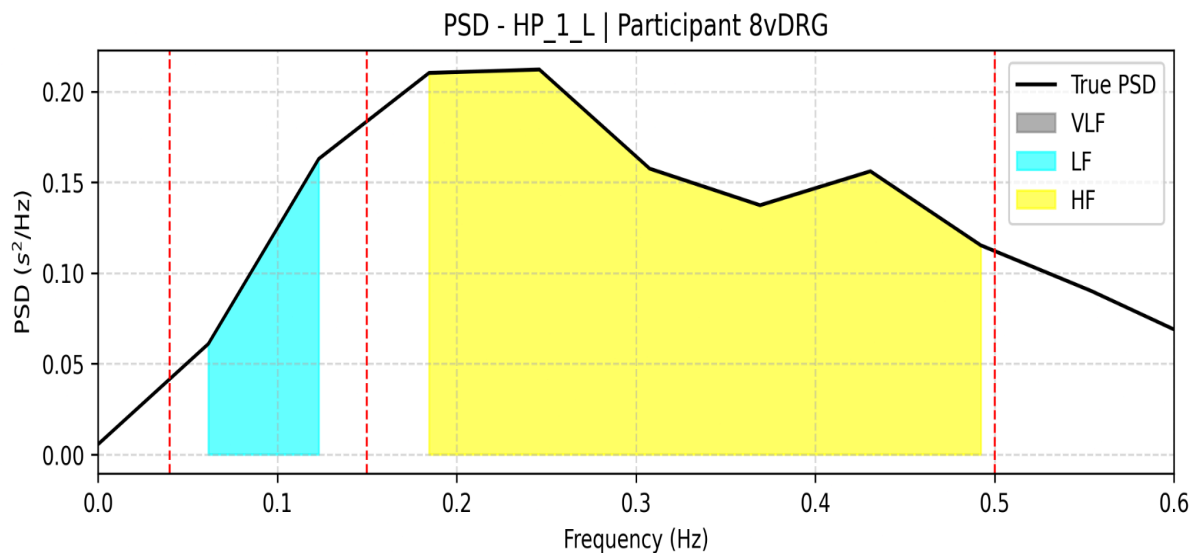


Figure 1.9 PSD plot HP_1_L, Participant 8vDRG

While the participant 8vDRG was exposed to the video HP_1_L, the dominance was in the section HF, which happens to show a parasympathetic modulation, which makes sense since it is a video tagged with an L of Low Arousal.

Non-linear features:

To capture more complex dynamics, Poincaré plots were generated, plotting each RR interval against the subsequent one. From these plots, features such as SD1 (short-term variability), SD2 (long-term variability), SD1/SD2 ratio, crossing points, elliptical area, and higher-order descriptors like skewness and kurtosis were derived. These non-linear measures provide insights into the peculiar properties of the cardiac system, which linear methods may miss.

The aim of this stage is to comprehensively represent the variability in cardiac activity under different emotional stimuli, ensuring that both linear and non-linear dynamics are preserved for subsequent feature selection and modeling.

Poincaré plots

As mentioned before Poincaré Plots are two dimensional scatter plots where each RR interval is plotted against the next one (RR_n vs. RR_{n+1}). They provide a simple yet powerful visualization of the nonlinear dynamics of heart rate variability.

- A tight, elongated ellipse along the line of identity suggests low variability, while a more spread out cloud indicates higher variability.
- Quantitatively, measures such as SD1 (shortterm variability, width of the ellipse) and SD2 (long term variability, length of the ellipse) are extracted.
- Additional descriptors such as the SD1/SD2 ratio, area of the ellipse, crossing points, and higher-order statistics (skewness, kurtosis) enrich the feature set.

Examples

Low arousal (stimuli HP_1_L)

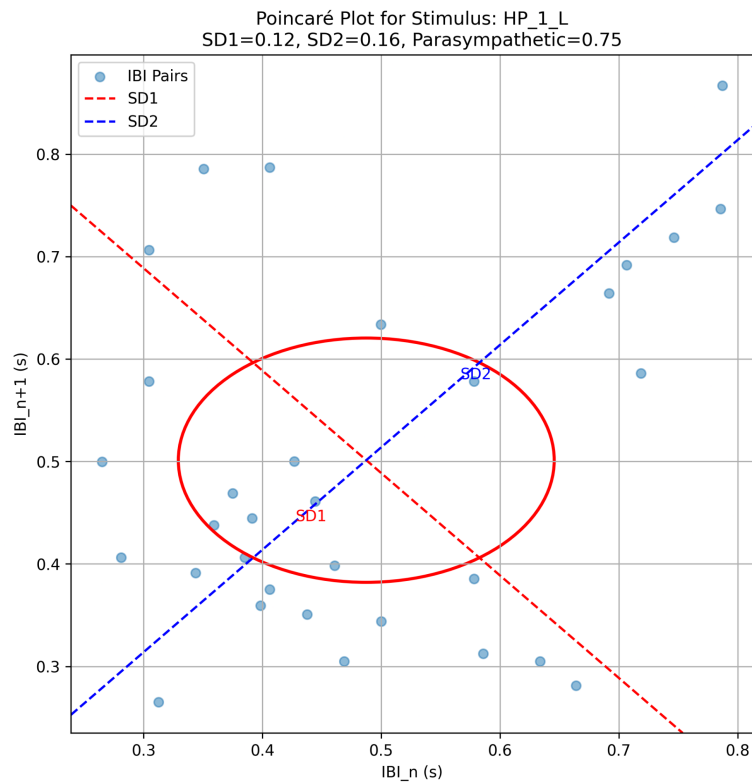


Figure 2.0 Poincaré plot, Participant 5BJD3, HP_1_L

This plot represents a low arousal state by having a wider ellipse with high values of SD1 (0.12) and SD2 (0.16). These parameters show greater short term (SD1) and long term (SD2) variability in the IBIs. In other words, higher HRV in this context suggests a dominant parasympathetic influence (Parasympathetic Index = 0.75). The wide dispersion of IBI pairs across the plot emphasizes the adaptability of the cardiovascular system, as the heart is able to fluctuate more freely in response to less strong emotions.

From a physiological perspective, such HRV behavior is typical of relaxed states, where the parasympathetic nervous system modulates cardiac activity to maintain homeostasis. This contrasts with high arousal conditions, where sympathetic dominance usually reduces variability and produces a narrower ellipse.

High arousal (stimuli HP_3_H)

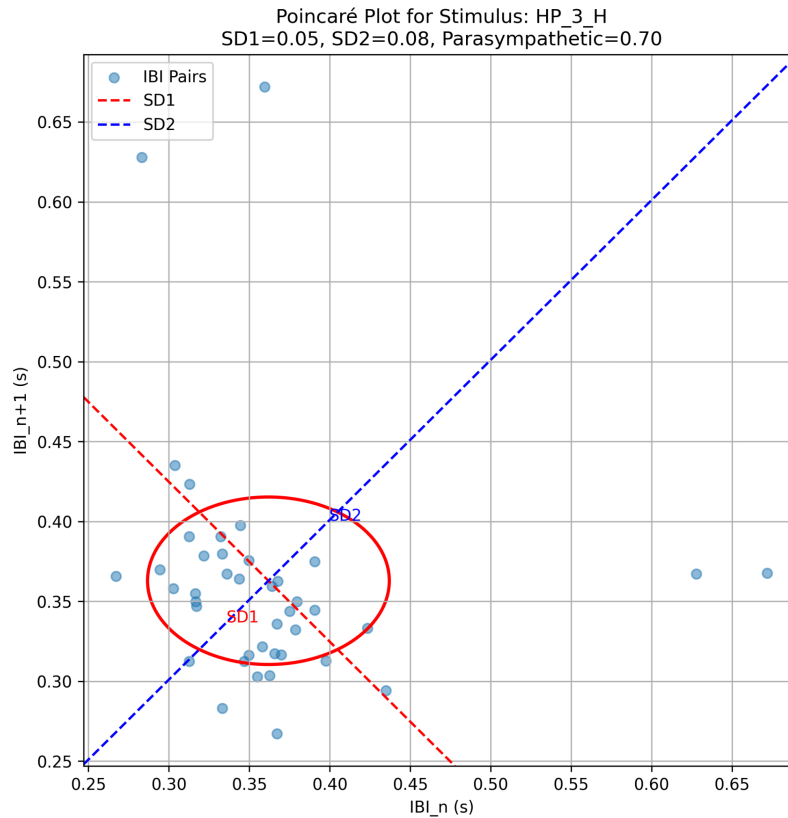


Figure 2.1 Poincaré plot, Participant M4t7k, HP_3_H

On the other hand, the high arousal state can be seen as in Figure 2.1, where the high arousal is represented by a tighter and more clustered ellipse, with noticeably lower values of SD1 (0.05) and SD2 (0.08). These reduced indices reflect diminished short term and long term variability in the interbeat intervals, indicating that the cardiovascular system is operating in a more constrained manner. From a physiological view, this suggests a dominance of sympathetic activity, which typically arises under stress, excitement, or heightened alertness, while parasympathetic modulation is suppressed.

The compact distribution of points in the plot demonstrates reduced flexibility of the autonomic nervous system to adapt to the stimuli. That is a pattern consistent with increased physiological arousal, where the heart rhythm becomes more rigid and less variable. This reduction in HRV is a recognized biomarker of stress and heightened emotional engagement.

Neutral calibration condition

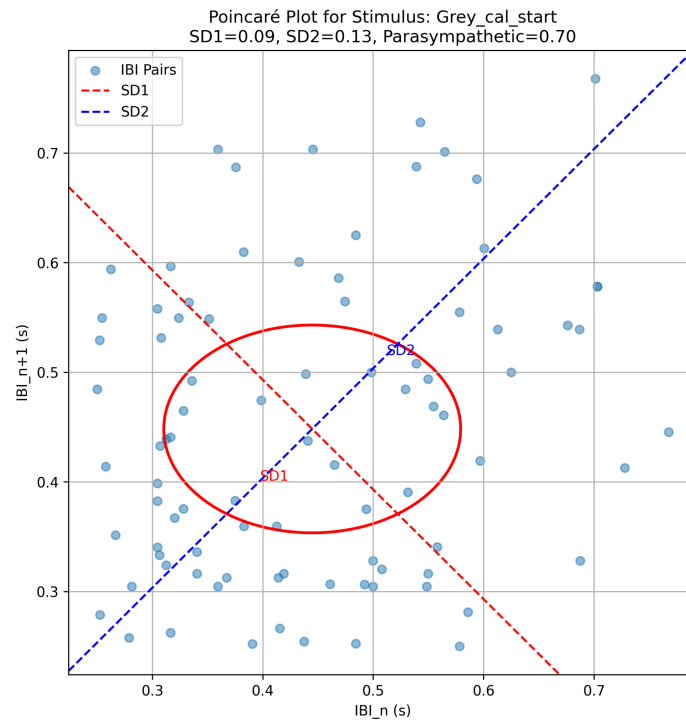


Figure 2.2 Poincaré plot, ParticipantERqc8, Grey_cal_start

The Poincaré plot for the Grey_cal_start (Figure 2.2) stimulus shows a relatively wide dispersion of inter beat interval pairs, represented by the ellipse formed around the points. The short term variability ($SD1 = 0.09$) and long-term variability ($SD2 = 0.13$) are both moderate, producing a parasympathetic index ($SD1/SD2$) of 0.70. This balanced ratio indicates healthy autonomic regulation, with contributions from both sympathetic and parasympathetic branches. The spread of points suggests that the heart is not locked into a rigid rhythm but instead demonstrates adaptability, which is expected in a calibration or neutral condition. This pattern reflects a stable physiological state without signs of either heightened arousal or excessive relaxation, making it a suitable baseline for normalizing subsequent HRV responses to emotional stimuli.

6. Feature Merging

Once the features from all domains (time, frequency, and non-linear) are extracted for each participant and stimulus, the next step is to merge them into a single dataset. Each row in this dataset corresponds to a participant stimulus pair, while the columns represent the extracted HRV features.

This merging process ensures that all feature sets (e.g., SDNN, RMSSD, LF/HF ratio, SD1, SD2) are aligned with the correct participant and stimulus, making it possible to systematically compare physiological responses across conditions. For example, time domain features provide information about beat to beat variability, frequency domain features capture oscillatory dynamics of autonomic control, and non-linear Poincaré metrics describe the geometrical structure of heart rate variability.

By combining them, the dataset captures a comprehensive representation of HRV for each stimulus, enabling more robust correlation analysis and further machine learning modeling.

7. Feature Normalization

To account for inter individual variability, all HRV features were normalized with a reference baseline. Different candidate baselines were initially tested, including some meditation and calibration phases. However, in order to ensure consistency, the choice of baseline was guided by seeing the IBI's plots to see the most stable one but mostly by comparing HRV stability indicators across baselines, particularly SDNN and RMSSD.

These metrics were used because:

- SDNN reflects overall variability in the signal; lower values indicate a more stable and less fluctuating heart rhythm.
- RMSSD captures short term variability driven by parasympathetic activity; lower values suggest reduced modulation and thus a steadier reference.

Following this criterion, the Grey_cal_start stimulus was selected as the optimal normalization reference. This baseline consistently exhibited the lowest SDNN and RMSSD values across participants, ensuring a stable and neutral physiological condition for comparison.

The normalization was applied to all extracted HRV features using the formula:

$$X_{norm} = \frac{X_{video} - X_{baseline}}{X_{baseline}}$$

Equation 1.5 Normalization formula

Where X_{video} is the feature value during a stimulus, and X_{baseline} is the feature value from `Grey_cal_start`. This approach transformed the features into relative changes from the baseline, expressed as proportional variations, making inter-participant comparisons more reliable.

8. Merging Features with Ground Truth Arousal

After normalization, the features dataframe was merged with the ground truth file which is based on the self reported arousal scores collected after each stimulus.

The join strategy implied a “left merge” in order to keep every row from the features table and bring the corresponding arousal values. This ensured that every participant-stimulus feature vector would receive its ground truth label when available.

Immediately after the merge, the missing values were checked so the rows with arousal = NaN would be noticed and printed a list of the affected participants. These same ones were then removed to prevent training on unlabeled samples.

9. Correlation Analysis

Once the normalized HRV features were merged with the ground truth arousal ratings, the next step was to analyze the relationship between the physiological features and subjective emotional arousal. For each participant, all features (time, frequency, and non linear) were normalized between -1 and 1, and correlations were computed with the normalized arousal scores. This ensured that differences in scale across features and participants did not bias the results.

Scatter plots were then generated for each feature, showing the distribution of normalized feature values against normalized arousal levels, with the Pearson correlation coefficient (r) displayed. This allowed for a visual inspection of linear trends, while also providing a quantitative summary of how strongly each HRV feature tracked arousal across stimuli.

To facilitate interpretation, three outputs were created:

- Participant level normalized datasets, containing all features alongside arousal for each stimulus.
- A global dataset, concatenating all participant level files into a single CSV for pooled analysis.
- A correlation summary file, reporting the feature wise correlation coefficients for each participant.

This analysis provided an initial view of which HRV features were most consistently aligned with arousal responses across participants.

Example:

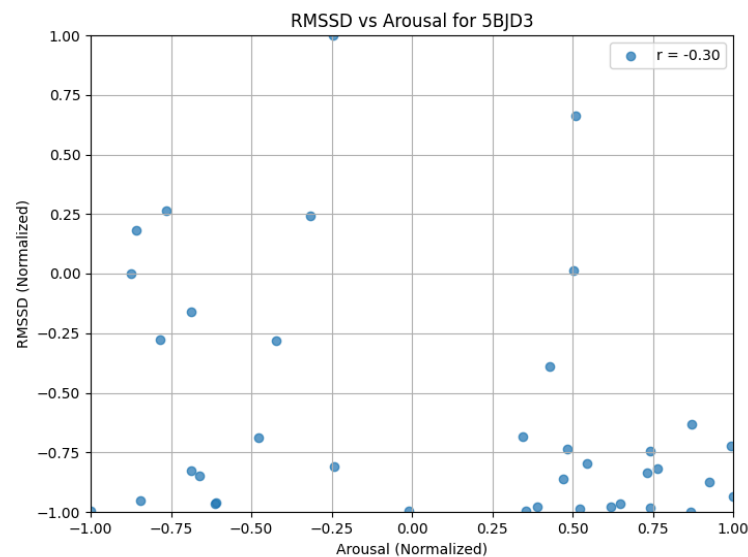


Figure 2.3 Correlation plot of feature RMSSD with the Arousal

These sort of plots were meant to visualize across the participants which features show most consistency in correlation with the arousal, this in order to see which are the ones most important to use for the Machine Learning models. With the summary created with the code, it was easy to visualize and count how many participants would show a good correlation between what features and the ground truth. This decision of “good” correlation was made by having a threshold parameter value of correlation of 0.40. To decide if to use a feature, the principle was that if most of the participants showed good correlation, then that feature would seem to look useful. However, this step to do the features selection was not very useful, since when doing the machine learning steps, the results were not very powerful.

That is why other approaches were considered, approaches which would analyze the data not only in a linear way, and that happens in the last code, where all the database with the features normalize and ground truth is used for doing the features selection and Machine Learning.

10. Dataset Preparation

- From the merged dataset (normalized HRV features + arousal ground truth), the predictor matrix X included the HRV derived features (time, frequency, non-linear, Poincaré).
- The target variable was Arousal, while Participant ID was retained for grouped validation.

11. Feature Selection

The feature selection was a principle thing to do to reduce dimensionality and retain only the informative predictors:

Two methods were chosen:

- Correlation analysis: The correlation was computed to have it as a reference and see how strong the features-ground truth arousal were.

```
F1_CrossingPoints  -0.351969
pVLF               -0.151748
aVLF               -0.109718
pLF                0.100511
F3_MaxArea        -0.086162
nLF                0.071986
F9_KurtX           -0.060054
aTotal             -0.053120
nHF                -0.046888
NN50               -0.045010
dtype: float64
```

Figure 2.4 Global Correlations scores by ascending order.

Nevertheless, as mentioned before a no-linear relation was explored to see if it would improve the further results.

- Mutual Information (MI): Computed the dependency between each feature and arousal, capturing both linear and non-linear associations. Features with $MI > 0.02$ were selected, resulting in a reduced set including the following:

Note: The parameter threshold of 0.02 was selected as it was the one which gave the best final results.

- RMMSD
- NN50
- pNN50
- aVLF
- aHF
- aTotal
- pVLF
- pLF
- nLF
- nHF
- F1_CrossingPoints

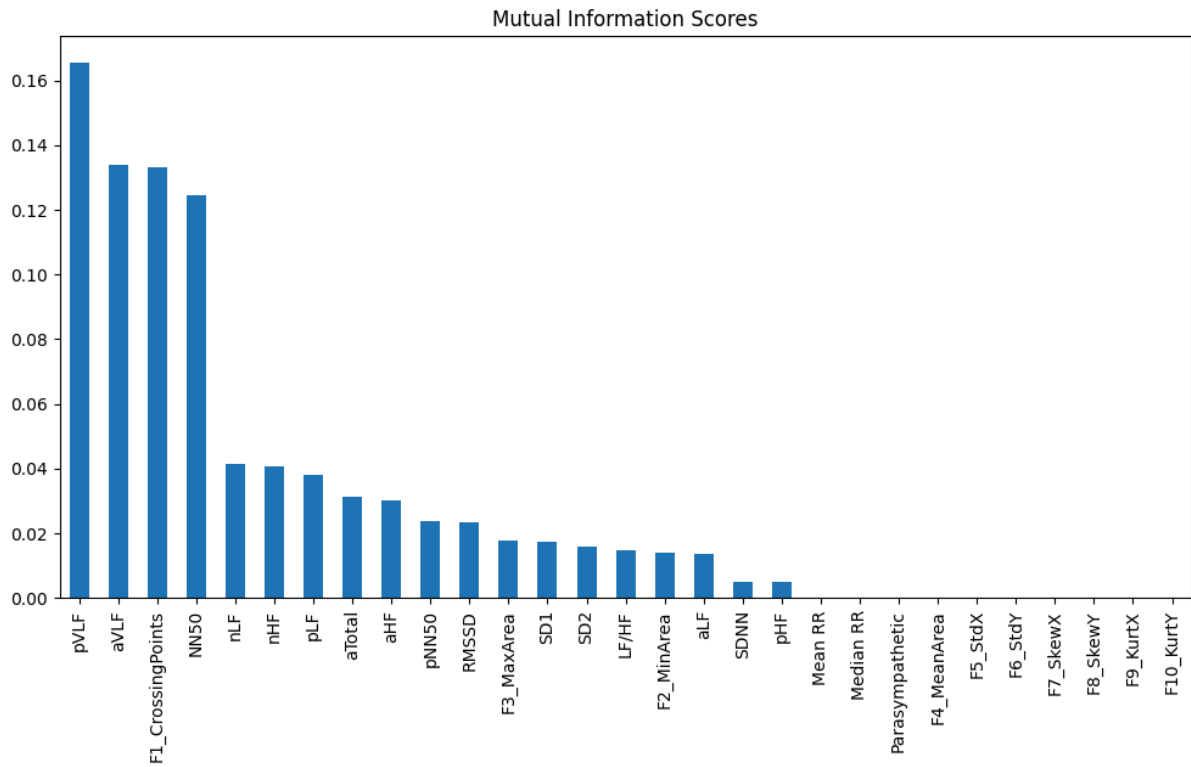


Figure 2.5 Mutual Information Scores

pVLF	0.165426
aVLF	0.133885
F1_CrossingPoints	0.133037
NN50	0.124604
nLF	0.041493
nHF	0.040712
pLF	0.037955
aTotal	0.031338
aHF	0.030152
pNN50	0.023663
RMSSD	0.023284
F3_MaxArea	0.017775
SD1	0.017523
SD2	0.015862
LF/HF	0.014821
F2_MinArea	0.014177
aLF	0.013822
SDNN	0.004924
pHF	0.004887
Mean RR	0.000000
dtype: float64	

Figure 2.6 Mutual Information Scores in Ascending Order

12. Model training and validation

For the model training and validation, three models were chosen based on their complementary strengths, their proven performance in physiological signal analysis, and their sustainability for datasets of the size and complexity used in this study.

Algorithms evaluated:

- XGBoost Regressor

The eXtreme Gradient Boosting known as XGBoost, was considered as it is a very powerful and widely used machine learning algorithm for structured data problems, especially those involving regression and classification. Among many reasons, XGBoost is very popular due the fact that it is designed not only to be very accurate but also to be efficient in terms of computation and scalability. This made it a good candidate to use for this project of emotion recognition using HRV features, where the dataset contains many features, participants, and stimuli, and the model needs to capture complex non-linear relationships between them.

The idea behind this XGBoost Regressor consists of an ensemble learning method. Instead of relying on a single model, this method builds a collection of decision trees, each one making predictions that complement and correct the mistakes of the previous ones. This process is known as boosting. More specifically, this regressor uses gradient boosting, which means that the model improves itself step by step by minimizing the errors (residuals) of the previous predictions using gradient descent techniques. Each new tree is built to reduce the difference between the predicted values and the actual labels, gradually making the overall model more accurate[29].

One of the key advantages is its capability on how it manages and balances bias and variance. Simple models, like a single decision tree, tend to have high bias (they cannot capture the complexity of the data), while very complex models may have high variance (they overfit to the training data and fail to generalize). XGBoost finds a balance by combining many relatively simple trees, focusing on improving the predictions aiming for a strong predictive model.

Moreover an important thing from the XGBoost is that it includes several built-in mechanisms to improve performance and prevent overfitting, those are the following:

- Regularization: it applies penalties to the complexity of the trees like the number of branches and leaf weights. This discourages the model from becoming too complex and ensures that it focuses on the most relevant patterns [29].

- Handling missing data: It can naturally deal with missing values by learning the best path for them during training[29]. This is useful in physiological data, where certain data may not always be available.
- Parallelization and efficiency: Unlike traditional boosting methods, this one is optimized for speed. It has the capability to train models much faster thanks to its ability to parallel computation, efficient use of memory, and advanced techniques like tree pruning and weighted quantile sketch[30]. This makes it possible to handle larger datasets without compromising performance.

For regression tasks like the one of this project, predicting arousal, XGBoost seemed to be a very good option to consider due its big probability of being effective because it can capture non-linear interactions between features and as it is known, emotional responses are rarely simple or linear. HRV features such as time-domain measures, frequency-domain components, and non-linear dynamics interact in complex ways depending on multiple factors such as in the individual and the stimulus. XGBoost, through its sequential tree-building boosting process, is capable of learning these subtle patterns better than linear models.

In summary, XGBoost is an advanced boosting algorithm that combines decision trees into a strong and efficient predictive system. Its robustness, scalability, and ability to handle noisy or incomplete data make it a particularly appropriate choice for modeling complex physiological responses and that is why it was considered to see how well it would predict the arousal from PPG-derived HRV features.

- Random Forest Regressor

The Random Forest Regressor is an ensemble learning algorithm that builds on the idea of combining multiple decision trees to make more robust and accurate predictions. While a single decision tree is simple and easy to interpret, it has the disadvantage of being very sensitive to the specific data it is trained on [31]. Therefore, the Random Forest solves this problem by creating a large collection, or “forest”, of decision trees and then averaging their predictions. This approach reduces overfitting and variance, leading to a much more stable model.

The construction of a Random Forest involves two main concepts: bagging (bootstrap aggregating) and feature randomness.

Bagging means that each decision tree in the forest is trained on a different random sample of the training data, created by sampling with replacement[32]. This ensures that no single tree sees all the data, and each tree learns slightly different relationships.

Feature randomness is introduced by letting each tree consider only a random subset of the features when splitting a node[31]. This increases diversity among the trees, as different trees will focus on different subsets of HRV features.

When it comes to regression tasks such as in this case when predicting arousal levels, Random Forest works by taking the average prediction across all the trees in the forest. The idea is that while one tree might make mistakes in certain areas, another tree will likely correct those mistakes, and the averaging process would produce a more balanced and reliable outcome.

An important advantage of Random Forest is that it can also capture non-linear relationships and feature interactions without requiring extensive preprocessing. For HRV-based emotion recognition, this seems particularly valuable because physiological signals are as mentioned before often complex and influenced by multiple interacting factors. The Random Forest Regressor can automatically model these interactions without the need for explicitly defining them.

From a computational perspective, Random Forest is less efficient than XGBoost in very large datasets but remains practical and fast enough for medium-scale problems such as for the dataset in this project. It also has fewer hyperparameters to tune compared to boosting algorithms, which makes it relatively more straightforward to use.

In summary, the Random Forest Regressor is a robust, interpretable, and flexible model that tackles the weaknesses of individual decision trees by averaging the results of many. Its ability to handle noisy data and capture complex non-linear relationships relevant features makes it an appropriate choice for predicting emotional arousal from HRV features derived from PPG signals [31].

- Support Vector Regressor (SVR, RBF kernel)

The Support Vector Regressor (SVR) is basically an adaptation of the Support Vector Machine, but instead of separating data into classes it is used for predicting continuous values. That makes it a good option here, since the goal of this project was not to classify emotions into fixed categories but to estimate arousal levels on a scale.

The main idea is that SVR does not try to perfectly match every single data point. Instead, it defines a kind of tolerance zone, often called the epsilon-tube. Predictions that fall inside this zone are considered “good enough” and do not get penalized, while only the larger deviations matter. This way, the model is less sensitive to small fluctuations, which is useful because PPG and HRV signals naturally contain little ups and downs that are not always meaningful.

Only a subset of the training points, the support vectors, really define the regression function. These are the points that sit on the edge or outside of the tolerance tube, and they are the ones that guide the final shape of the model. In practice, this makes SVR fairly compact and efficient, because it does not need to rely equally on every single observation.

One of the biggest strengths of SVR is that it can use kernel functions to capture non-linear relationships. The RBF kernel (Radial Basis Function) is especially common and powerful: it basically allows the model to map the original HRV features into a higher-dimensional space where non-linear patterns become easier to capture. This is valuable in HRV analysis, since the interaction between sympathetic and parasympathetic activity is rarely linear.

SVR also has parameters that control how flexible or strict the model is. The parameter C adjusts the balance between fitting the training data closely versus keeping the model smoother and more general. A very large C means the model will try hard to reduce training error, sometimes at the cost of overfitting. On the other hand, smaller values of C lead to a more general model. The γ (gamma) parameter, which applies to the RBF kernel, decides how far the influence of a single data point reaches. Lower values of γ give broader influence, producing smoother functions, while higher values of γ can capture finer details but may also lead to noise being learned as signal. Finally, ϵ (epsilon) controls the width of the tolerance tube mentioned earlier.

In this project, these parameters were tuned carefully because the HRV features varied a lot between participants, and the signals contained both noise and individual differences. Compared to the other models, Random Forest and XGBoost, SVR can perform quite well when the dataset is not very large but the feature space is high dimensional. Its main drawback is that training can get heavy on larger datasets, but here the data size was manageable, so this was not a limitation.

Overall, the SVR with an RBF kernel provided a way of testing whether a flexible, non-linear approach could map the relationship between HRV features and emotional arousal. Its ability to tolerate small noise while still reacting to larger deviations made it a natural candidate for this kind of physiological data.

Evaluating methods

Cross-validation strategy:

- Leave-One-Participant-Out (LOPO) was adopted to simulate generalization to unseen individuals. Since inter individual variability in HRV is high, LOPO validation was used to ensure the models generalize across participants rather than overfitting individual patterns.
- Nested 5-fold inner CV was used for hyperparameter tuning.

By testing these three algorithms separately, the aim was not to combine them into a single model, but rather to evaluate which approach best captured the relationship between HRV features and arousal. This comparative approach also ensures that the analysis is not biased toward a single modeling framework, but instead explores the performance of diverse regression strategies before concluding on their suitability for this physiological emotion recognition approach.

Evaluation Metrics

Models were assessed using the following metrics:

- RMSE, MAE, R^2
- Pearson correlation (r) with p-values
- Concordance Correlation Coefficient (CCC) to measure agreement beyond correlation.

Metrics chosen because they are standard in regression tasks with physiological signals.

13. Results

The evaluation of the three regression models: XGBoost, Random Forest, and Support Vector Regressor (SVR with RBF kernel) was carried out in two stages. First, baseline results were obtained using default parameters to establish an initial reference point. Then, hyperparameter optimization was applied through nested cross-validation to assess whether more careful tuning could significantly improve predictive performance.

This two step approach was chosen because in physiological signal analysis, model performance can vary widely depending on parameter settings. Baseline performance offers insight into the raw ability of each model to capture the relationship between HRV features and arousal, while the tuned results demonstrate better potential of each algorithm when adjusted to the dataset characteristics.

The results are presented separately for each model, showing the differences between the untuned and tuned versions in terms of the chosen evaluation metrics (RMSE, MAE, R^2 , Pearson correlation, p-value, and Concordance Correlation Coefficient). This structure also allows for a fair comparison across the algorithms and provides insights into their strengths and limitations in the context of HRV based emotion recognition.

XGBoost Regressor

- The model achieved an RMSE of 0.169 ± 0.032 , MAE of 0.132 ± 0.028 , R^2 of 0.218 ± 0.207 , Pearson correlation of $r = 0.5145$, P-value of $4.7390e-139$ and CCC of 0.4896 . These results indicated that XGBoost was already capable of capturing non-linear patterns between HRV features and arousal, though with modest explained variance.

- With hyperparameter tuning:
Performance improved, with RMSE reduced to 0.154 ± 0.034 , MAE to 0.128 ± 0.033 , and R^2 increased to 0.350 ± 0.092 . Correlation also rose to $r = 0.5975$ as the P-value to $7.2221e-199$ with a CCC of 0.4851.
This shows that careful adjustment of parameters such as learning rate, max depth, and subsample size allowed the model to generalize more effectively across participants.

Random Forest Regressor

- Without hyperparameter tuning:
The Random Forest Regressor achieved RMSE of 0.156 ± 0.033 , MAE of 0.125 ± 0.033 , R^2 of 0.340 ± 0.115 , correlation of $r = 0.5839$, and CCC of 0.5197.
Notably, even without tuning, this model outperformed baseline XGBoost and SVR in terms of explained variance and agreement (CCC).
- With hyperparameter tuning:
After parameter optimization (max depth, n_estimators, and minimum samples per split/leaf), Random Forest further improved, reaching RMSE of 0.154 ± 0.032 , MAE of 0.122 ± 0.033 , and R^2 of 0.357 ± 0.125 , with correlation $r = 0.5975$ and CCC 0.5334.
These results demonstrate that Random Forest consistently provided robust and stable predictions, confirming the best suitability for HRV-based arousal prediction.

Support Vector Regressor (SVR, RBF kernel)

- Without hyperparameter tuning:
The baseline SVR achieved RMSE of 0.170 ± 0.039 , MAE of 0.139 ± 0.037 , R^2 of 0.207 ± 0.162 , correlation of $r = 0.4602$, and CCC of 0.3808.
This was notably lower than the ensemble based methods, suggesting limited capacity to capture the full complexity of the dataset without parameter adjustment.
- With hyperparameter tuning:
The optimization of C and γ values did not lead to significant improvements. Results remained similar: RMSE of 0.170 ± 0.038 , MAE of 0.139 ± 0.036 , R^2 of 0.209 ± 0.155 , correlation of $r = 0.4619$, and CCC of 0.3814.
This indicates that while SVR can model some non-linear dynamics, it is less competitive than tree based approaches for this specific dataset.

Model	Configuration	RMSE	MAE	R^2	Correlation (r)	CCC
XGBoost	Baseline	0.169 ± 0.032	0.132 ± 0.028	0.218 ± 0.207	0.5145	0.4896

	Tuned	0.154 ± 0.034	0.128 ± 0.033	0.350 ± 0.092	0.5975	0.4851
Random Forest	Baseline	0.156 ± 0.033	0.125 ± 0.033	0.340 ± 0.115	0.5839	0.5197
	Tuned	0.154 ± 0.032	0.122 ± 0.033	0.357 ± 0.125	0.5975	0.5334
SVR (RBF)	Baseline	0.170 ± 0.039	0.139 ± 0.037	0.207 ± 0.162	0.4602	0.3808
	Tuned	0.170 ± 0.038	0.139 ± 0.036	0.209 ± 0.155	0.4619	0.3814

Table 2.1 Results

Results Analysis

Overall, the tree-based ensemble methods (XGBoost and Random Forest) consistently outperformed SVR across most evaluation metrics. Both XGBoost and Random Forest showed better ability to capture the non-linear interactions present in HRV data, reflected in higher correlation values and lower RMSE compared to SVR. This is consistent with the expectation that physiological signals such as HRV exhibit non-linear dynamics, which can be difficult for models like SVR to fully capture when the dataset is noisy and relatively small.

Hyperparameter Tuning Impact

When comparing baseline models with tuned versions, a clear improvement was observed across all three algorithms. Hyperparameter optimization significantly reduced error (MAE and RMSE) and improved correlation scores. For example, Random Forest demonstrated notable gains in correlation after tuning, reinforcing the importance of carefully adjusting model parameters rather than relying on default configurations.

This result underlines that while baseline models provide an initial benchmark, tuning is essential in physiological data modeling where noise and inter-subject variability are high.

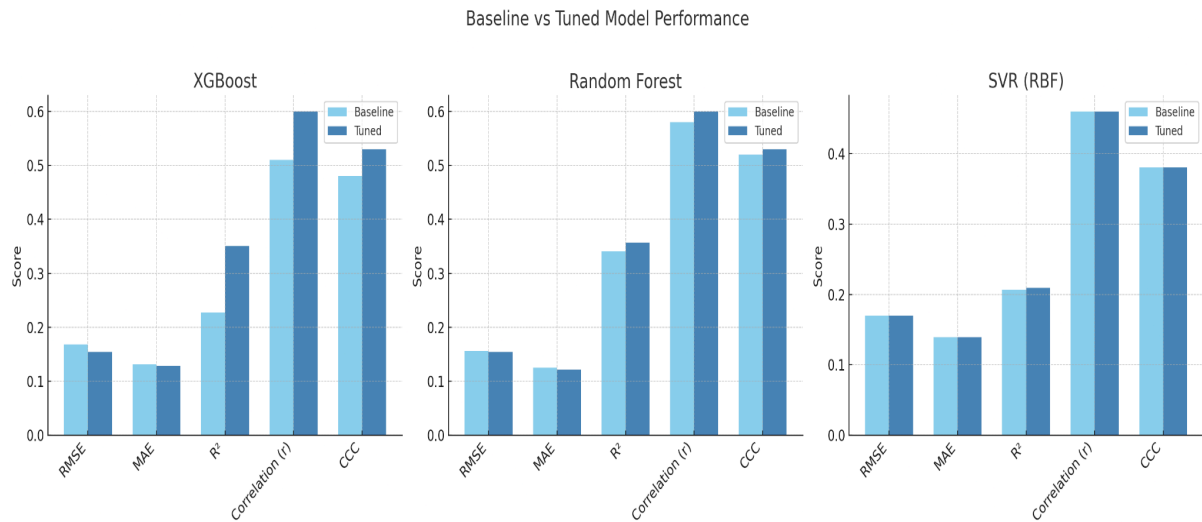


Figure 2.7 Baselines vs Tuned Model Performance

Best Performing model

Among the tested models, Random Forest emerged as the most reliable and best performing algorithm, achieving the highest correlation values and lowest error rates after hyperparameter tuning. XGBoost also performed competitively, showing strong results; however, the Random Forest slightly surpassed it. On the other hand, SVR, while theoretically suited for non-linear data, was less stable and underperformed compared to the ensemble methods.

Comparison with Literature

The results obtained in this work can be connected with what has been seen in previous studies that used PPG derived HRV features for emotion recognition. A lot of the existing work has focused more on classification, like Rakshit et al. (2016), who reported accuracies above 80% when separating emotions such as happy, sad or neutral using SVMs. In this case, the task was different because the aim was continuous arousal prediction, which is a more demanding approach. The lower correlations found here (around 0.59 at best) reflect this difference, since regression needs to capture small changes in arousal on a continuous scale, while classification only separates broader emotional categories.

Another important reference comes from the use of non-linear HRV features. For example, Goshvarpour and Goshvarpour (2018) showed very high accuracies (above 90%) by using Poincaré measures to classify emotions while participants watched videos. Even if that study was classification based, it still supports the decision of using non-linear metrics in this project, as these features were also relevant here and helped the performance when predicting arousal.

Looking only at regression based approaches, Silva et al. (2022) showed that HRV estimation from PPG alone is possible, but they reported a correlation coefficient of only 0.40, which

underlines the challenges of noise and variability in PPG. The results from this project (0.59 correlation) suggest that with the right preprocessing, normalization and feature extraction, it is possible to achieve higher and more stable outcomes.

Finally, Schidt et al. (2018), through the WESAD dataset, reported R^2 values around 0.64 for arousal prediction with LSTMs. While that project achieved better results, this project showed that working with these other more traditional models such as Random Forest and XGBoost can still reach competitive performance.

In few words, compared with previous studies, the results here confirm that PPG derived HRV features can be used for continuous arousal prediction, reaching performance levels that are in line with or better than earlier regression benchmarks. At the same time, it shows that while multimodal and deep learning approaches can get slightly higher results, PPG alone with well-structured preprocessing and machine learning could be already a reliable tool for emotion recognition.

Physiological Interpretation

The fact that ensemble models performed better suggests that emotional arousal is not linearly tied to a single HRV parameter, but rather to a combination of features capturing time-domain variability (e.g., SDNN, RMSSD), frequency-domain shifts (e.g., LF/HF ratio), and non-linear dynamics (e.g., Poincaré indices). This supports the idea that arousal is a multi dimensional physiological response, requiring models capable of integrating diverse inputs rather than relying on simplified assumptions.

Conclusion

The purpose of this work was to explore whether heart rate variability, time, domain, and non-linear features extracted from PPG signals could be useful to predict emotional arousal through machine learning. To achieve this, a full pipeline was developed, starting with PPG preprocessing, extraction of time, frequency, and non-linear HRV features, normalization, merging with arousal labels, and then applying regression models.

The extraction of the features and their analysis showed a broad perspective on how rate responds to multiple stimuli. Patterns were clear: high-arousal states generally produced lower variability in beat to beat intervals, while relaxed states showed higher variability. Visual tools like PSD plots and Poincaré plots supported this by illustrating the differences between conditions. Among all the features: RMSSD, NN50, pNN50, aVLF, aHF, aTotal, pVLF, pLF, nLF, nHF, F1_CrossingPoints stood out as strong markers of arousal.

On the modeling side, the results showed that flexible algorithms like Random Forest and XGBoost were good at performing, achieving stronger correlations with the arousal and lower error values. SVR also captured non-linear interactions but was much less consistent.

Beyond the numbers, the study contributes a complete pipeline: from preprocessing and feature extraction, to normalization, relation analysis, and model evaluation. This not only provides a reproducible framework but also highlights the potential of using a single, non-invasive signal like PPG for emotion recognition or other important uses. This work showed that while PPG based HRV is less precise than multimodal setups, it can be enhanced as a promising option for scalable and practical applications.

Before looking forward, it is important to identify the troubles and barriers that occurred in the process, which were : basically in the nature of the raw data, in some participants too much data was missing, therefore, the imputation techniques were applied to get the best guesses; however, having good sensor lectures at every moment would be the best. Besides that, another very important detail was the duration of the intervals and the content of the stimuli, if the shorts were larger and would show more aroused content, the analysis could be improved since more important data could be extracted and analyzed to distinguish the physiological changes or better said in this case the Heart Rate Variability which as said before requires of some amount of seconds to analyze properly since it has a delay to demonstrate reaction and its neutralization is not instantaneous. It is common to see that most of the projects involving the analysis of heart variability consider a lecture time of around more than 150 seconds, while in this project some of the intervals are only 15 seconds length, being that one of the reasons the modelling might have not performed as well as in other circumstances would. Moreover, it is relevant to consider the length of the database which could be more useful when capturing more participants data, so the bigger it could get the more opportunity the models can have to find patterns and perform better.

Future research should work with more complete and larger datasets, explore deep learning models for sequential HRV data, and test the system in real world environments such as neuromarketing or adaptive media. These steps would help bring PPG based emotion recognition closer to practical applications.

In conclusion, this study shows that HRV features derived from PPG contain meaningful information about emotional arousal, and that machine learning models can leverage this to

provide reliable predictions. It demonstrates both the challenges and the promise of building emotion recognition systems from a simple, wearable friendly signal.

Appendix

All project codes:

https://github.com/defeolab/Multimotion_application/tree/hr_Diego

https://github.com/zp20945/Group-project-CE903_HR

https://github.com/DArmandoSalinas/HR_DiegoSalinas/tree/main

All project analysis (All codes results):

Links:

Box: <https://essexuniversity.box.com/s/10nrvu0u2t09tx3fkd0jqvkqeo9bzelj>

GoogleDrive:  Diego Armando Salinas

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